ExpNo 16 Experiment to understand EDA Quantitative and Qualitative Analysis

Program:

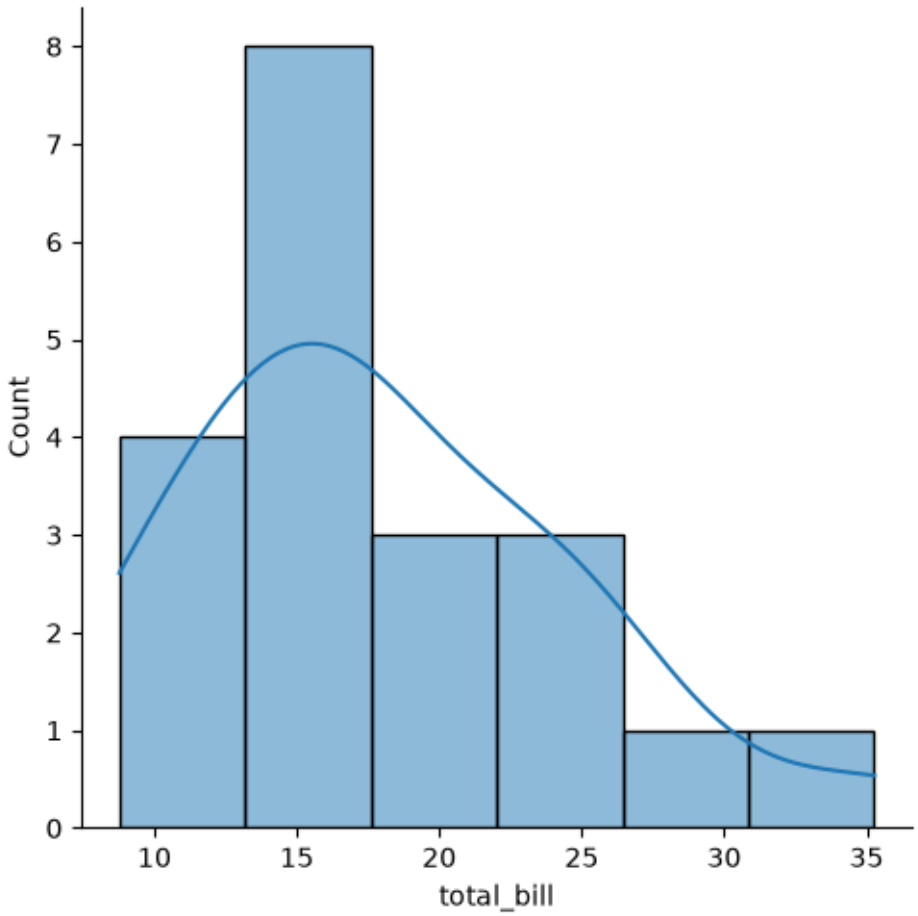
```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
tips=pd.read_csv('tips_modified.csv')
tips.head()
sns.displot(tips.total_bill,kde=True)
sns.displot(tips.total_bill,kde=False)
sns.jointplot(x=tips.tip,y=tips.total_bill)
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
sns.pairplot(tips)
tips.time.value_counts()
sns.pairplot(tips,hue='time')
sns.pairplot(tips,hue='day')
sns.heatmap(tips.corr(numeric_only=True),annot=True)
sns.boxplot(tips.total_bill)
sns.boxplot(tips.tip)
sns.countplot(x=tips.day)
sns.countplot(x=tips.sex)
```

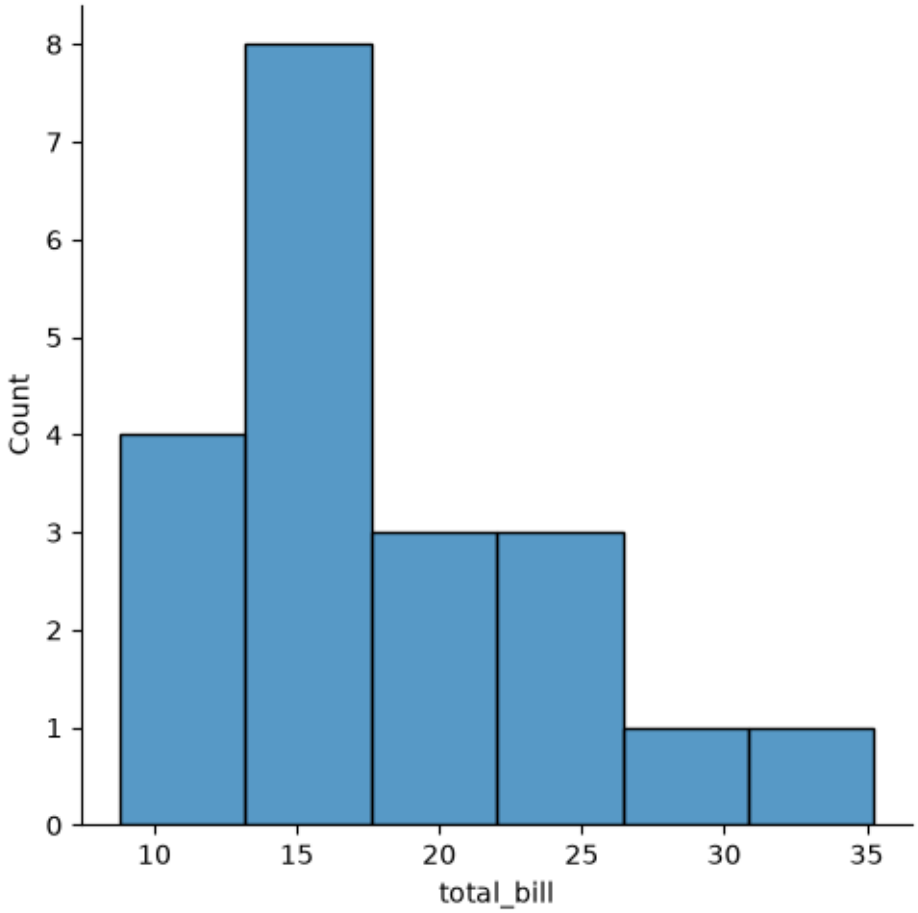
tips.sex.value_counts().plot(kind='pie')

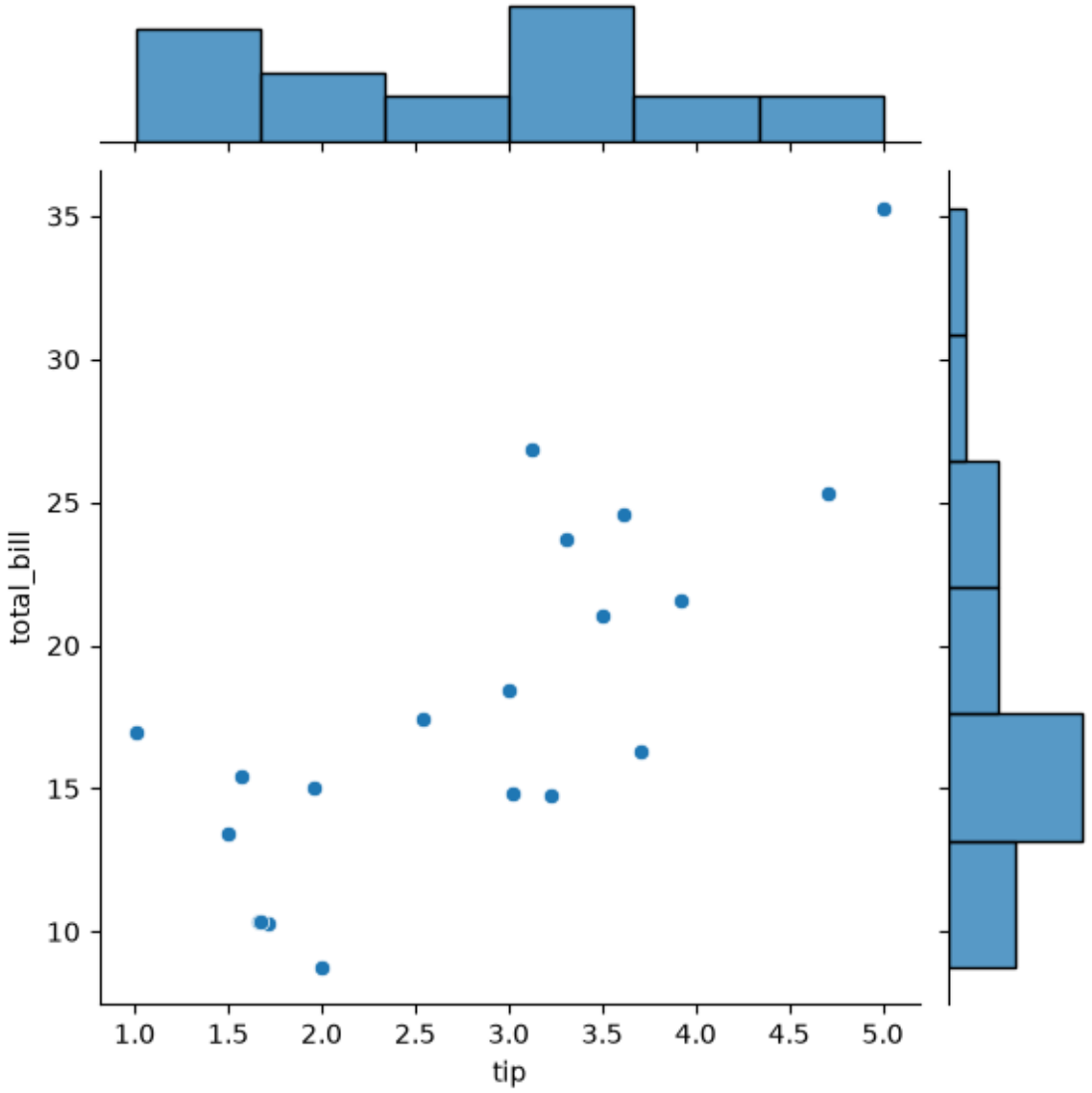
tips.sex.value_counts().plot(kind='bar')

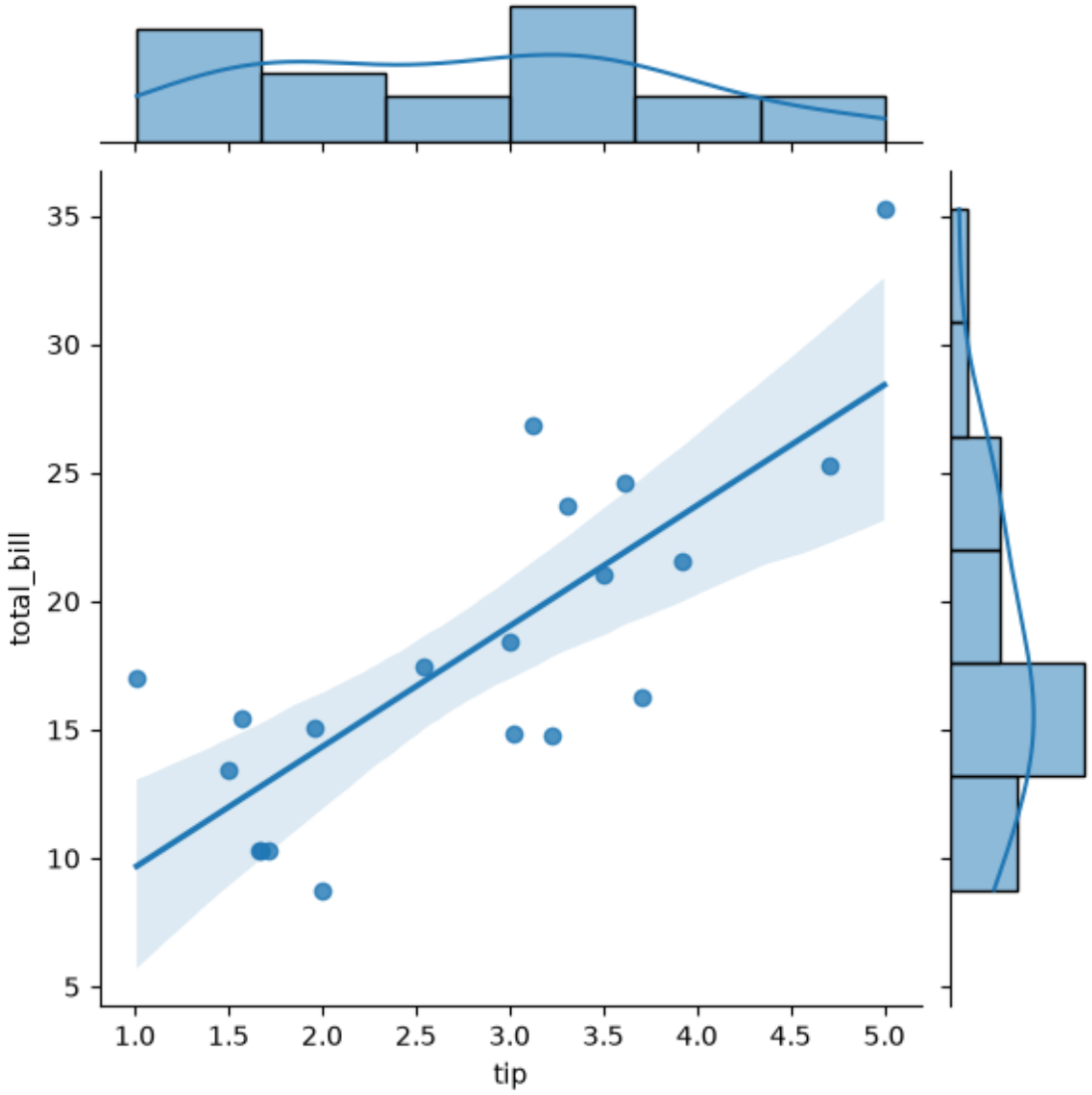
sns.countplot(x=tips[tips.time=='Dinner']['day'])

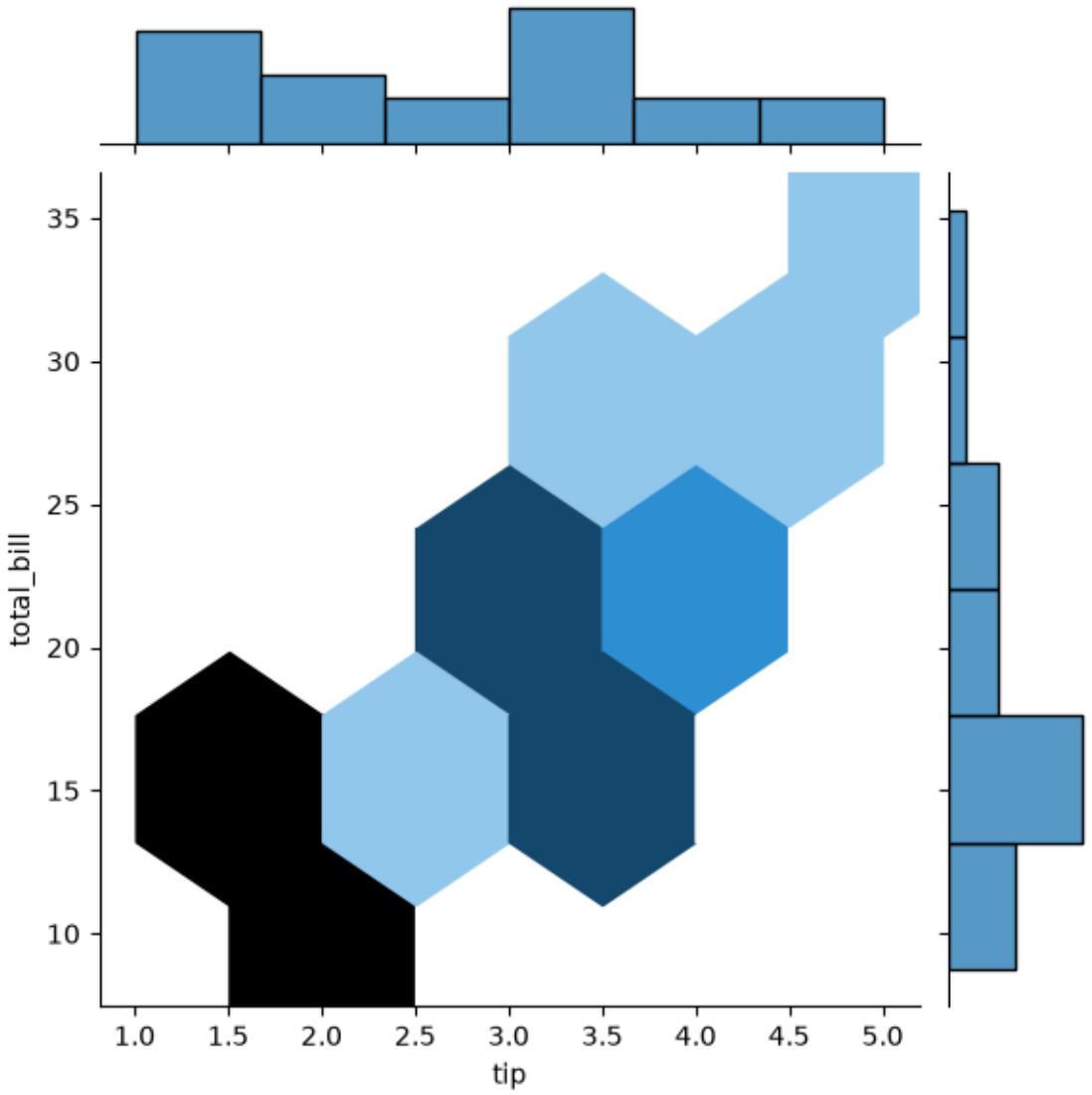
Output:

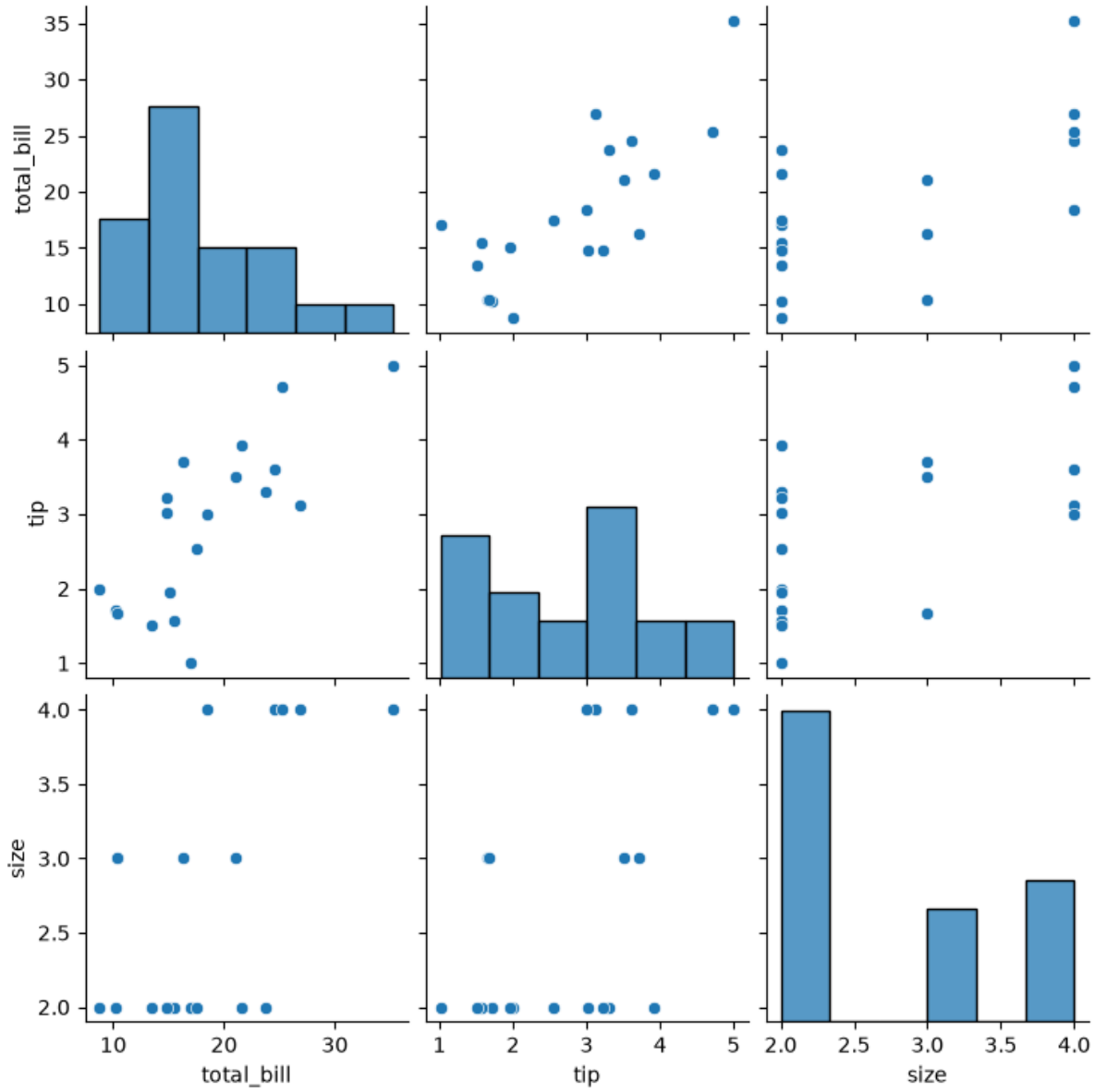


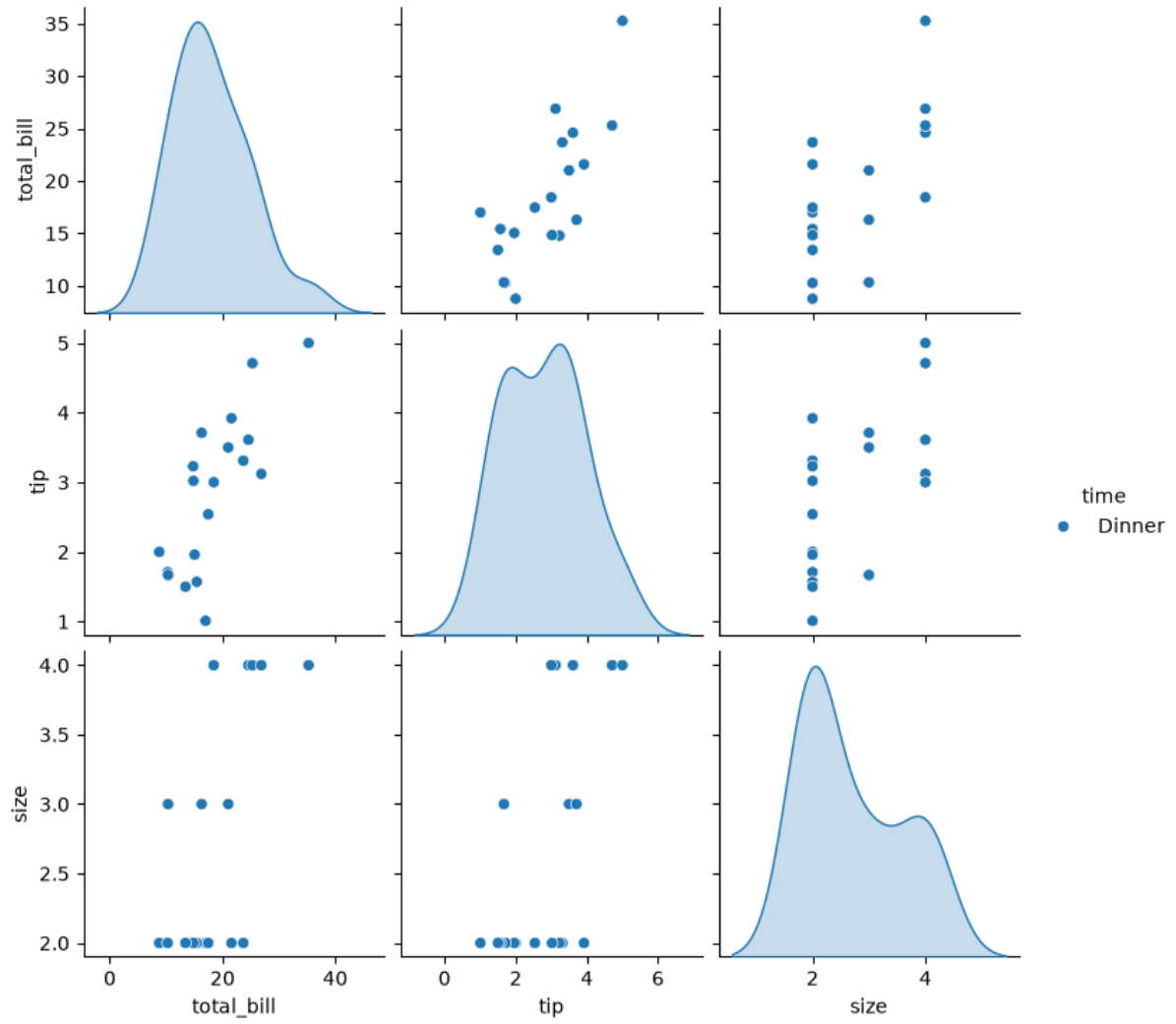

Count
0
1
2
3
4
5
6
7
8
10
15
20
25
30
35
total_bill

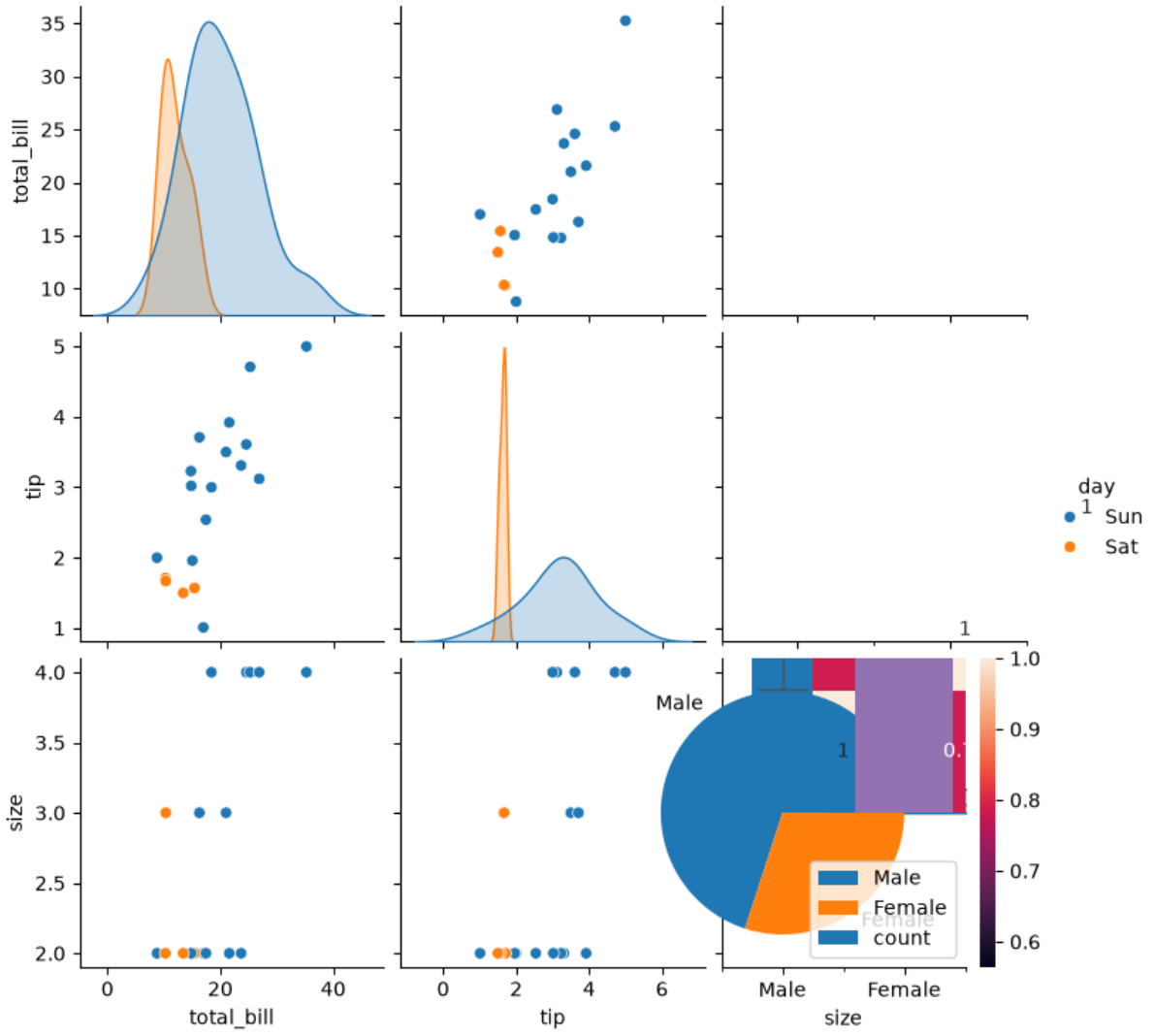

35
30
25
20
15
10
total_bill
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
tip


35
30
25
20
15
10
5
total_bill
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
tip


35
30
25
20
15
10
total_bill
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
tip


total_bill
tip
size
10
20
30
1
2
3
4
5
2.0
2.5
3.0
3.5
4.0


total_bill
tip
size
time
Dinner


total_bill
tip
size
day
Sun
Sat
Male
Female
count
Male
Female
total_bill
tip
size